VU Quiz Collection

Total Quizzes: 10

Courses: MTH603 - Numerical Analysis (Quiz No 02)

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Quiz 1 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

Gauss - Jordan Method is similar to …….

Options:

A. Relaxation Method

B. Gaussian elimination method

C. Iteration’s method

D. Gauss–Seidel method

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Quiz 2 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

Let [A] be a 3x3 real symmetric matrix with |a23||a23|\left| {{a_{23}}} \right| be numerically the largest off-diagonal element of A, then we can construct orthogonal matrix S1 by Jacobi’s method as

Options:

A. #¡#£#¢1000cos¸sin¸0" sin¸cos¸#¤#¦#¥[1000cosa¸" sina¸0sina¸cosa¸]\left[{\begi
1&0&0\\0&{\cos \theta }&{ - \sin \theta }\\0&{\sin \theta }&{\cos \theta }
\end{array}} \right]

B. #¡#£#¢1000cos¸sin¸0" cos¸" sin¸#¤#¦#¥[1000cosa¸" cosa¸0sina¸" sina¸]\left[{\beg
1&0&0\\0&{\cos \theta }&{ - \cos \theta }\\0&{\sin \theta }&{ - \sin \theta }
\end{array}} \right]

C. #¡#£#¢cos¸sin¸0" sin¸cos¸0001#¤#¦#¥[cosa¸" sina¸0sina¸cosa¸0001]\left[{\begi
{\cos \theta }&{ - \sin \theta }&0\\{\sin \theta }&{\cos \theta }&0\\0&0&1
\end{array}} \right]

D. #¡#£#¢cos¸0sin¸010" sin¸0cos¸#¤#¦#¥[cosa¸0" sina¸010sina¸0cosa¸]\left[{\begi
{\cos \theta }&0&{ - \sin \theta }\\0&1&0\\{\sin \theta }&0&{\cos \theta }
\end{array}} \right]

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Quiz 3 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

A 3 x 3 identity matrix have three and _____eigen values.

Options:

A. different

B. same

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Quiz 4 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

Iterative algorithms can be more rapid than direct methods.

Options:

A. TRUE

B. FALSE

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Quiz 5 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its main diagonal.

Options:

A. FALSE

B. TRUE

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Quiz 6 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

For two matrices A and B, such that “A = Inverse of B”, then which of the following is true?

Options:

A. AB = Identity Matrix

B. All choices are true

C. A and B are non-singular

D. B = Inverse of A

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Quiz 7 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

While using Jacobi method for the matrix
A =#¡#£#¢ 1 1 / 4 1 / 2 1 / 4 1 / 3 1 / 4 1 / 2 1 / 4 1 / 5#¤#¦#¥ A = [1 1 / 4 1 / 2 1 / 4 1 / 3 1 / 4 1 / 2 1 / 4 1 / 5] A = \left[
1&{1/4}&{1/2}\\{1/4}&{1/3}&{1/4}\\{1/2}&{1/4}&{1/5}
\end{array}} \right]
the value of ‘theta ¸’ can be found as

Options:

A. tan2¸=2a13a11" a33tana2¸=2a13a11" a33\tan 2\theta = \frac{{2{a_{13}}}}{

B. tan2¸=2a13a33" a11tana2¸=2a13a33" a11\tan 2\theta = \frac{{2{a_{13}}}}{

C. tan2¸=2a12a11" a22tana2¸=2a12a11" a22\tan 2\theta = \frac{{2{a_{12}}}}{

D. tan2¸=2a23a22" a33tana2¸=2a23a22" a33\tan 2\theta = \frac{{2{a_{23}}}}{

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Quiz 8 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

Power method is applicable if the eigen vectors corresponding to eigen values are linearly independent.

Options:

A. FALSE

B. TRUE

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Quiz 9 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

Eigenvalues of a symmetric matrix are all _____.

Options:

A. positive

B. negative

C. real

D. zero

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Quiz 10 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

If the product of two matrices is an identity matrices that is AB = I, then which of the following is true?

Options:

A. A is inverse of B

B. B is singular

C. A is transpose of B

D. A is singular

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